

1 Functions

1.1 Introduction

Perhaps the most important mathematical object to mathematicians and people who apply mathematics is the notorious *function*. It is something we can use to describe (as accurately or inaccurately as possible) how things, relationships, or processes *function*. For example, we could, say, model the income of some business by the rule, *If we sell x things at a price p per x things, then our income is px* . This rule can be abstracted to what we call a *function*. To do so, we first name our function with a letter, say I . That is, I is the function (as a noun) that describes how the relationship between x and income function (as a verb) together under the rule described above. To indicate that I does this, we use some notational tools: We shall write $I(x)$ to be the value of income if x things were sold. That is, $I(x)$ is the quantity px .

Now, it is clear we can write $I(x) = px$ as the quantity $I(x)$ equals px and in fact this is how we usually describe a given function; viz., given a function f that describes how some set of things (the “ x ”) relates to another set of things (the “ $f(x)$ ”), we write $f(x) = \text{something}$, where *something* is the relationship or rule between the two somethings (i.e., the x and the $f(x)$, respectively). That is, the $f(x)$ depends on the x , which is why we call the x the **dependent variable**. We then call the first “set of things” (the set of x ’s) as the domain of the function f , and the “another set of things” (the set of $f(x)$ ’s) as the codomain of the function f . Moreover, for some more notational convenience, we name the domain and codomain of function by letters. So, one may say, the domain of a function f is D and the codomain is R , where these names are arbitrary (viz., we could have called the domain L for all we care).

Now that we have seen some introductory remarks about functions, it is time to formalize this concept with definitions and notations that will be used throughout these notes.

1.2 Some Set Theory

What exactly *is* set theory will not worry us here. We care only about the notational usage of set theory (whatever that is) and we shall list this usage here. So let’s see our first definition!

Definition 1.2.1. By a **set** A , we mean any (abstract) collection of (abstract) objects.

Note that we are again naming things rather arbitrarily; however, when I say, for example, the domain of a function f is A , we always understand (in context) that A *is* the domain of f and nothing else. Moreover, we are speaking rather abstractly. However, we shall always work with numbers and thus our sets will always be collections of numbers (e.g., the set of even numbers, the set of real numbers, the set of rational numbers).

There are several ways to indicate the elements (which is what we call the things that a set contains) of a set. But, since we shall usually focus on what are called intervals, we will mainly focus on two notations, the first being interval notation and the second being inequality notation. Let’s see inequality notation first.

Definition 1.2.2. Let a and b be any real numbers. Then, when we say A is the set of x such that $a < x < b$, we mean the set of numbers x that are greater than a and less than b .

Example 1.2.3. Let A be the set of x such that $1 < x < 2$. Then, $1.5, 1.2, \pi - 2, 1.2345$, etc., are all elements of the set A , and there are many more! Note that 1 nor 2 are in A .

We often want to consider sets that are unbounded in either the negative or the positive side; i.e., we want to say that elements of some set A can become arbitrarily negative or positive, respectively. To do this, we include the notation $-\infty$ or $+\infty$ (the latter often written by dropping the ‘+’ sign and writing ∞), to mean the following. The notation $-\infty$ is to mean that which is always more negative than any number; i.e., all real numbers x are such that $-\infty < x$. The notation ∞ is to mean that which is always more positive than any number; i.e., all real numbers x are such that $x < \infty$.

Example 1.2.4. Let A be the set of x such that $-\infty < x < 5$. Then, all $x < 5$ is include in A since all real numbers are greater than $-\infty$.

The next notation we consider is so called interval notation.

Definition 1.2.5. Let a and b be any reals number. Then, if we say $A = (a, b)$, we mean A is the set of x such that $a < x < b$.

Example 1.2.6. Let $A = (3, 5)$. Then, $3.1, \pi, 4, 4.9$, etc., are all elements of the set A , and there are many more! Note that 3 nor 5 are in A .

Now, we may construct a grand table that depicts these definitions and extensions of the definitions to include more cases

Interval Notation	Inequality Notation	Explanation
$a < x < b$	(a, b)	All x greater than a and less than b
$a < x \leq b$	$(a, b]$	All x greater than a and less than or equal to b
$a < x < \infty$ or $a < x$	(a, ∞)	All x greater than a
$a \leq x < b$	$[a, b)$	All x greater than or equal to a and less than b
$a \leq x \leq b$	$[a, b]$	All x greater than or equal to a and less than or equal to b
$a \leq x < \infty$ or $a \leq x$	$[a, \infty)$	All x greater than or equal to a
$-\infty < x < b$ or $x < b$	$(-\infty, b)$	All x less than b
$-\infty < x \leq b$ or $x \leq b$	$(-\infty, b]$	All x less than or equal to b
$-\infty < x < \infty$	$(-\infty, \infty)$	All x

Now, for one last piece of notational usage, we speak of unioning two sets.

Definition 1.2.7. Let A and B be two sets. Then we write $A \cup B$ to mean the set that contains elements of both A and B .

Example 1.2.8. Let $A = [0, 1]$ and $B = (1, 2)$. Then $A \cup B = [0, 2)$.

Example 1.2.9. Let $A = [1, 2]$ and B be the set of x such that $3 < x < 4$. Then $A \cup B = [1, 2] \cup (3, 4)$. Note that we cannot combine the interval notations as we did above, and this is okay and common practice.

Note that understanding both of notations and unioning will be very important for later material.

1.3 Functions

We are now ready to handle functions more mathematically and in a way we will need to for the rest of the material. Let's finally define *what* a function *is*.

Definition 1.3.1. Let D and R be sets. Then f is a **function** from D to R if and only if to each D , f is a rule that assigns one and only one element in R . Mover, D is called the **domain** and R the **codomain**

Example 1.3.2. Let $D = (-\infty, \infty)$ and $R = (-\infty, \infty)$. Then f defined by the equation $f(x) = x^2$ is a function from D to R . This is so because for each x in D , f assigns the number x^2 to x ; i.e., f assigns one and only one element in R to each D .

So what does f actually do here? We have defined f by the rule $f(x) = x^2$; i.e., f assigns to each number x , its square. So, for example, $f(2) = 2^2 = 4$, $f(3.3) = 3.3^2 = 10.89$, and so on.

In this example, note that f does not assign any x in D to any element of $(-\infty, 0)$, and this is okay! The definition of a function does not require the function to assign everything in the codomain. This brings up the next definition.

Definition 1.3.3. Let f be a function with domain D and codomain R . Then the set y such that $f(x) = y$ for some x in D is called the **range** of f ; i.e., it is the set of all numbers for which f assigns elements of D to. Put another way, for inputs x in D , the range is the set of outputs $f(x)$.

Let's see few more examples.

Example 1.3.4. Let g be the function whose domain is $[0, 2]$, whose codomain is $(-\infty, \infty)$, and is defined by $g(x) = x^3$. Thus g assigns to each number in $[0, 2]$ its cube. For example, $g(0) = 0$, $g(1) = 1^3 = 1$, and $g(2)^3 = 8$. Moreover, the range of g is $[0, 8]$ as can be reasoned out—how we can show this rigorously shall be covered later, but a sort of “hand wavy” explanation/understanding will be sufficient.

Example 1.3.5. Let h be the function whose domain is $[0, 10]$, whose codomain is $(-\infty, \infty)$, and is defined by $h(x) = x^3$ for $0 \leq x \leq 2$ and $h(x) = x$ for $2 \leq x \leq 10$. Thus for any x in $[0, 2]$, h assigns the cube of x and for any x in $(2, 10]$, h assigns x itself. So, for example, $h(1.5) = 1.5^3 = 3.375$ and $h(5) = 5$. Lastly, the range of h is the range of h on $[0, 2]$ unionized with the range of h on $[2, 10]$; i.e., the range of h is $[0, 8] \cup [2, 10] = [0, 10]$.

1.4 A Convention

Before we go on, you might have noticed that explicitly writing the domain and codomain of a function is tedious, which is why we employ the following convention from now on,

Convention. Suppose we are given a function f without explicitly mentioning its domain nor codomain. We shall then always assume that the domain of f is the largest possible set for which f makes sense and that its codomain is $(-\infty, \infty)$, unless otherwise specified. Note that this says nothing about the range of f . $\square$

What this means is that if we are given a function (by a graph or by an equation), we are to assume the domain of the function is the largest set that the function makes sense on.

Example 1.4.1. Suppose we are given the function f defined by the rule $f(x) = \frac{1}{x}$. Then, we *must* assume that the domain of f is $(-\infty, 0) \cup (0, \infty)$ since we cannot divide by zero. That is, this set is the largest possible set for which f makes sense.

Example 1.4.2. Suppose we are given the function g defined by the rule $g(x) = \sqrt{x}$. Then, we *must* assume that the domain of g is the set of x such that $0 \leq x$, since the square root of a negative number is not allowed. That is, this set is the largest possible set for which g makes sense.

Let’s see some three more difficult examples.

Example 1.4.3. Let $f(x) = \frac{1+x}{\sqrt{x+1}}$. We wish to find the domain of this function and so we wish to find where the function is undefined. We note that the $1+x$ in the numerator causes us no issue, but the $\sqrt{x+1}$ in the denominator potentially does. Firstly, the square root of a negative number is not defined and so we must firstly have $x+1 \geq 0$. Therefore, at the very least, $x \geq -1$. Moreover, we cannot divide by zero and so we must have $\sqrt{x+1} \neq 0$, and so, since $x+1 = 0$ at $x = -1$, we must have $x \neq -1$. Using these two results, we may conclude $x > -1$ or $(-1, \infty)$ is the domain of f .

Example 1.4.4. Let $f(x) = \frac{x+1}{(x^2+2x+1)}$. We wish to find the domain of this function and so we wish to find where the function is undefined. As above, $x+1$ in the numerator causes no issue, but the x^2+2x+1 in the denominator potentially does. What might occur? Well, if $x^2+2x+1 = 0$, we would be dividing by zero, and therefore, we need to determine where x^2+2x+1 does not equal to zero. So we factor, $x^2+2x+1 = (x+1)^2$; i.e., $x^2+2x+1 \neq 0$ so long as $x \neq -1$. Therefore, the domain of f is $(-\infty, -1) \cup (-1, \infty)$.

Example 1.4.5. Let $g(x) = \frac{x+2}{x^2+3x+2}$. As above, we only care where the denominator’s quadratic does not equal zero. Thus, we factor, $x^2+3x+2 = (x+1)(x+2)$. But then, $g(x) = \frac{x+2}{(x+1)(x+2)} = \frac{1}{x+1}$, right? Well, not quite! It turns out that g and the function $h = \frac{1}{x+1}$ are not quite the same. Consider what happens when we input -2 for x in g :

$$g(-2) = \frac{-2+2}{(-2)^2-6+2} = \frac{0}{0},$$

which is not defined! This may seem nit picky, and it is, but, at least by convention, we assume g and h are not the same function here. Therefore, even though $h(-2)$ makes sense, $g(-2)$ does not. We conclude that -2 and -1 are the only values for which g is undefined and therefore the domain of g is $(-\infty, -2) \cup (-2, -1) \cup (-1, \infty)$.

1.5 Manipulating Functions

The point of this section is see how one might manipulate and evaluate a given function, and thus enforce one's understanding of functions. The first example will be very important when defining derivatives, so please advise!

Example 1.5.1. Define the function f by $f(x) = \frac{1}{x+2}$. Note by the convention mentioned above that the domain of f is $(-\infty, -2) \cup (2, \infty)$ since only $f(-2)$ does not make sense. In any case, we will want to evaluate functions at values that are not explicitly chosen numbers. Say, for example, we want to see what $f(x+h)$ means. Well, in the same way we would input any old number, we just replace the 'x' in $f(x) = \frac{1}{x+2}$ exactly by ' $x+h$ ', giving us

$$f(x+h) = \frac{1}{x+h+2}.$$

I really want to stress that we have simply just replace 'x' by ' $x+h$ ' and that nothing fancy has occurred. (Strictly speaking, we need $x+h \neq -2$ since otherwise $f(x+h) = f(-2)$, which is not defined!)

Let's input some other values, including explicit and non-explicit numbers:

$$\begin{array}{lll} f(a) = \frac{1}{a+2} & f(a^2) = \frac{1}{a^2+2} & f(a+b) = \frac{1}{a+b+2} \\ f(x-h) = \frac{1}{x-h+2} & f(-x) = \frac{1}{-x+2} & f(x+2h) = \frac{1}{x+2h+2} \\ f(2) = \frac{1}{4} & f(4) = \frac{1}{6} & f(-3) = -1. \end{array}$$

The next example will show us how to add, subtract, multiply, and so on, functions.

Example 1.5.2. Let $f(x) = x^2$ and $g(x) = x+1$ be two functions. We note that we can understand what $f(x) \pm g(x)$, $\frac{f(x)}{g(x)}$, and $f(x)g(x)$ all mean by simply inputting for x and then doing the given operation. Let's see examples of this.

$$\begin{array}{lll} f(x) + g(x) = x^2 + x + 1 & f(2) + g(3) = 2^2 + 3 + 1 = 8 & f(-2) - g(-4) = (-2)^2 - (-4 + 1) = 7 \\ \frac{f(x)}{g(x)} = \frac{x^2}{x+1} & \frac{f(2)}{g(3)} = \frac{2^2}{3+1} = 1 & \frac{f(-2)}{g(-4)} = \frac{(-2)^2}{-4+1} = -\frac{4}{3} \\ f(x)g(x) = x^2(x+1) & f(2)g(3) = 2^2(3+1) = 16 & f(-2)g(-4) = (-2)^2(-4+1) = -12 \end{array}$$

Our last example will explore something really important for defining derivatives and builds what we have already seen just above.

Example 1.5.3. Let $f(x) = x^2 + 1$. The following computation will be really useful (I am really stressing this for a reason!) in the near future.

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 + 1 - x^2 - 1}{h} = \frac{x^2 + 2xh + h^2 - x^2 - 1}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Notice how the h has completely disappeared in the denominator! Take note again! it will be important.

1.6 Piecewise Functions

If you thought you were done with notation, you were wrong! Here we introduce one last piece of notation that will show up here and there; this notation being what is used to describe a piecewise function. We shall also see how to graph piecewise functions and how to model applications with piecewise functions.

In Example 1.3.5, we became aware that one need not require a function to have only one rule. In fact, we can construct a function with as many rules as we wish! We explore this idea by examples.

Example 1.6.1. Let us define a function f by the following two rules. On the set $[0, 1]$, let $f(x) = x+1$, and on $(1, 2)$, let $f(x) = x^3$. Note by the convention above, f has domain $[0, 1] \cup (1, 2) = [0, 2)$. We see that we have defined the function f by two separate rules: one for $[0, 1]$ and one for $(1, 2)$. To express this with convenience notation, we write the following

$$f(x) = \begin{cases} x+1 & 0 \leq x \leq 1 \\ x^3 & 1 < x < 2 \end{cases}$$

So how do we use this notation? We shall call each line of this piecewise function a **case**. So, say we want to compute $f(1.5)$. We then find for which case does 1.5 fit in, which is the second case since $1 < 1.5 < 2$. Therefore, to compute $f(1.5)$, we use case two: $f(1.5) = 1.5^3 = 3.375$. Similarly, if we want to compute $f(0.5)$, we use case one since $0 < 0.5 \leq 1$ and so $f(0.5) = 0.5 + 1 = 1.5$. As a last point for clarification, note that $f(1) = 1 + 1 = 2$ and not $f(1) = 1^3 = 1$ since $0 < 1 \leq 1$.

As an extra point, we have that the range of f is $[1, 2] \cup (1, 8) = [1, 8)$.

Example 1.6.2. Let us define a function g by the following three rules. On the set $[0, 1]$, let $f(x) = x + 1$; on $[1, 2]$, let $f(x) = 2x$; and on $[2, 3]$, let $f(x) = 4x$. Then, to express f as a piecewise function, we would write

$$f(x) = \begin{cases} x + 1 & 0 \leq x \leq 1 \\ 2x & 1 \leq x \leq 2 \\ 4x & 2 < x < 3. \end{cases}$$

In preparation for future material, let us consider two examples that deal with piecewise functions with at least one case that is defined at a single value rather than an interval.

Example 1.6.3. Let h be a function defined by the following two rules. On the set $(-\infty, 0) \cup (0, \infty)$, let $h(x) = \frac{1}{x}$; on the set containing only the number 0, let $h(x) = 1$. That is, h can be written as the piecewise function

$$h(x) = \begin{cases} 1 & x = 0 \\ \frac{1}{x} & \text{otherwise.} \end{cases}$$

Note that we can also write h as

$$h(x) = \begin{cases} \frac{1}{x} & x < 0 \\ 1 & x = 0 \\ \frac{1}{x} & x > 0. \end{cases}$$

Example 1.6.4. Let k be the function defined by the following two rule. On the set $(-\infty, 0) \cup (0, \infty)$, let $h(x) = \frac{1}{x}$. That is, h can be written as the piecewise function

$$k(x) = \begin{cases} \frac{1}{x} & x < 0 \\ \frac{1}{x} & x > 0. \end{cases}$$

This example shows us that it okay to define a piecewise function with a domain that is not $(-\infty, \infty)$; i.e., there may be “holes” in our function. Note that k in this example is different from h in the previous example.

1.7 The Absolute Value Function

The piecewise function considered in this section is important enough to warrant a name and will show up quite frequently. The function is called the **absolute value** function and is written as $f(x) = |x|$ (lo! another piece of notation!). This function is defined by the piecewise function

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0; \end{cases}$$

i.e., $|x|$ is always a positive number, equal in magnitude to x . For example, $|-2| = 2$, $|-5| = 5$, $|2| = 2$, and so on.

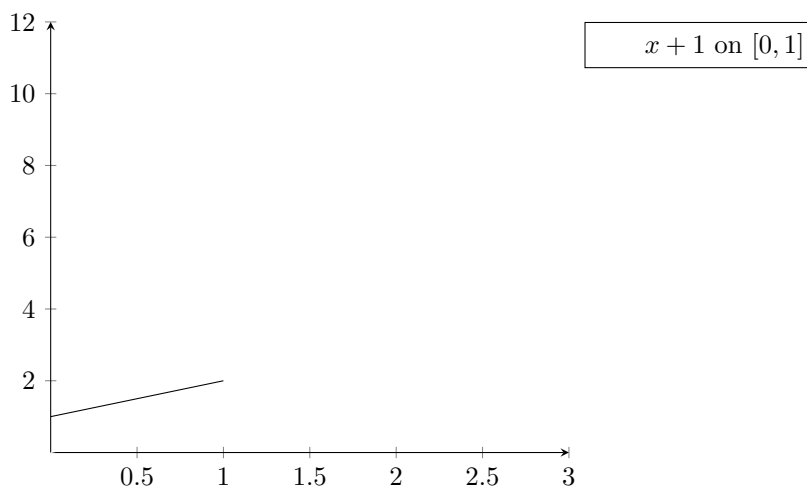
1.8 Graphing Piecewise Functions

To graph a piecewise function is essentially the way in which we normally graph a function, except now we graph each case of the piecewise function with respect to their respective intervals. There will be a convention to plotting piecewise graphs, however.

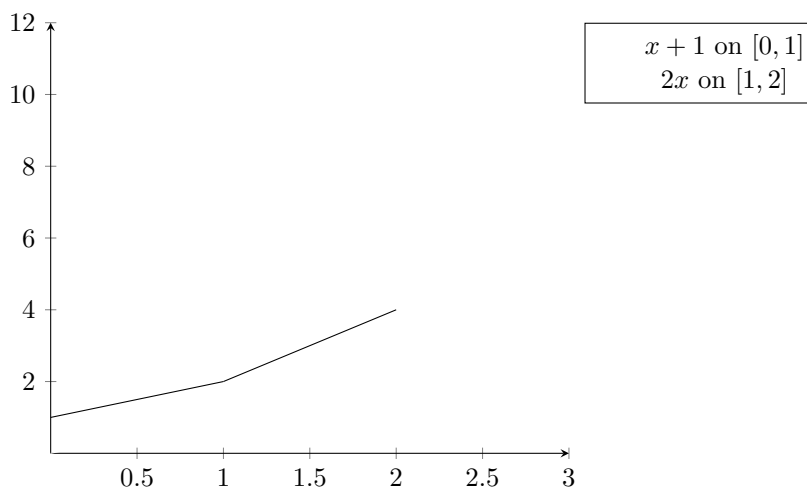
Convention. First draw bubbles at the end point of each case's domain. Second, if any cases are defined at any of these endpoints, fill in the bubble at such a point. $\square$

We shall show this and the method of graphing piecewise functions by example.

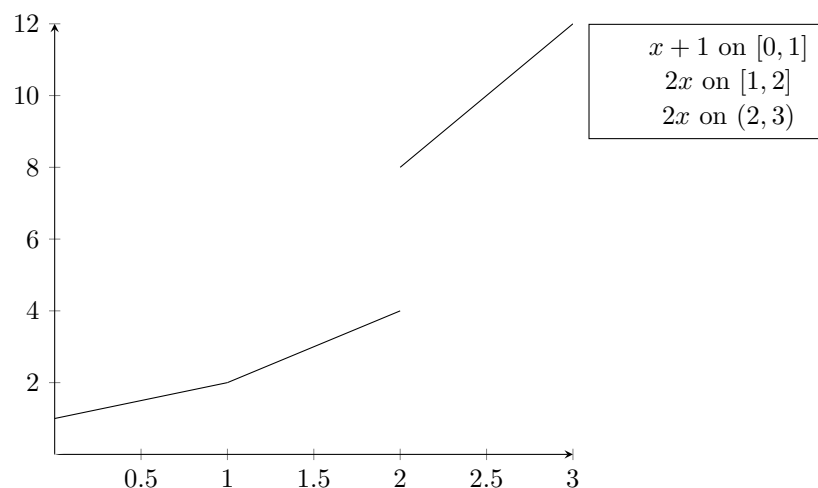
Example 1.8.1. Let h be the function in Example 1.6.2. We shall construct the graph of h step by step by considering each case of h and following the convention just mentioned. The first case of h is $x + 1$ on $[0, 1]$, so we must plot $x + 1$ on this interval.



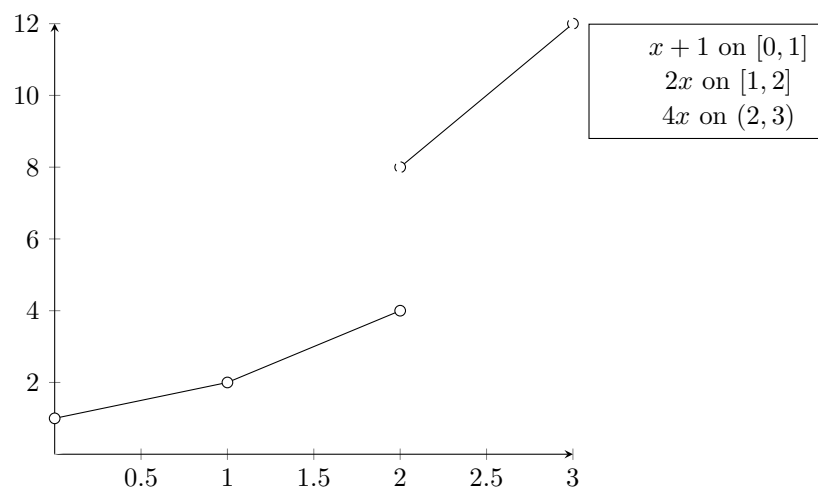
The second case of h is $2x$ on $[1, 2]$, so we must plot $2x$ on this interval.



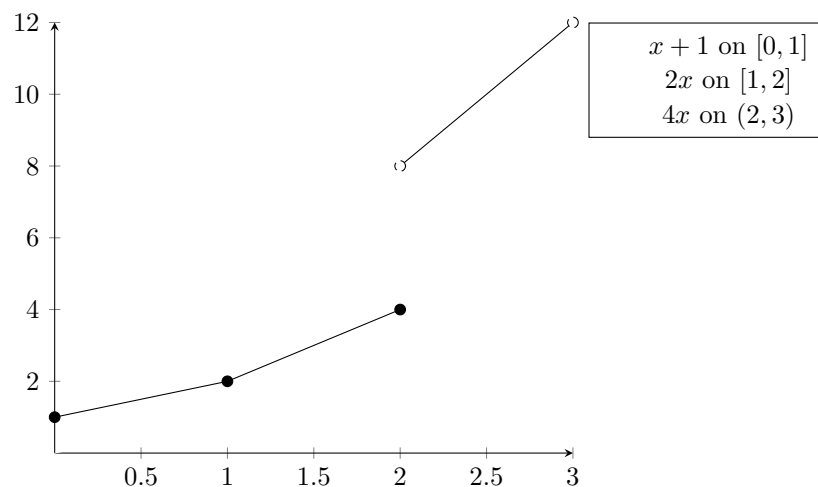
The last case of h is $4x$ on $(2, 3)$, so we must plot $4x$ on this interval.



We see that there is an ambiguity in these graphs with whether or not the points $(0, 1)$, $(1, 2)$, $(2, 4)$, $(2, 8)$, or $(3, 12)$. Thus we must apply the just mentioned convention. We first draw open bubbles at all of these points since 0, 1, 2 and 3 are end points of the cases' intervals.



But since $f(0)$, $f(1)$ and $f(2)$ are defined, and so we must fill in the corresponding bubbles $(0, f(0)) = (0, 1)$, $(1, f(1)) = (1, 2)$, and $(2, f(2)) = (2, 4)$. Moreover, since the case of f on $(2, 3)$ is not defined at 2 nor 3, we must leave these bubbles unfilled. We get the final graph.



1.9 Modeling with Piecewise Functions

The point of this section is to demonstrate how, when given a modeling type problem, how one can construct an appropriate piecewise function to model the situation.

Example 1.9.1. Suppose you run a business in need of some steel and that the steel supplier sells steel at \$800 per metric ton when purchasing up to 100 metric tons, and \$700 per metric ton when purchasing after 100 metric tons. Moreover, suppose the supplier requires you to pay a \$400 base service fee. Our task will be to model the cost of buying steel from this particular supplier.

Now, it is clear there are two cases to consider. If you wish to purchase steel at metric tons t in the interval $(0, 100]$, the steel will cost $\$800t + 400$, the \$800 coming from the price per metric tons, and the \$400 coming from the base service fee. Furthermore, if you wish to purchase steel at metrics tons t in the interval $(100, \infty)$, the steel will cost $\$700t + 400$. That is, if C is the cost function, then

$$C(t) = \begin{cases} 800t + 400 & 0 < t \leq 100 \\ 700t + 400 & 100 < t < \infty. \end{cases}$$